

REUBEN LAWRENCE | CS23B054

Computer Science and Engineering, IIT Madras

CGPA : 9.86  ScepticallySam  Reuben Lawrence

PROFESSIONAL EXPERIENCE

Software Engineering Intern

Optiver | Sydney, Australia May 18 – Jul 9, 2026

- Worked as a software engineering intern, contributing to software engineering projects at the trading firm.

Optiver Future Focus: Technology

Optiver | Mumbai, India Jun 16–20, 2025

- Learned how software engineering works at a trading firm through workshops and activities hosted by the Optiver team.

PROJECTS

StarPlat MLIR Migration

- Developed the StarPlat-to-MLIR translation layer, converting StarPlat programs into a structured MLIR representation and enabling integration with a multi-backend compilation pipeline supporting sequential CPU and parallel OpenMP execution.

Multi-Client Chat Server

- Developed a C++ TCP-based multi-client chat system with user authentication, broadcast and private messaging, active-user tracking, and persistent chat history.
- Implemented three server architectures using process-based, thread-based, and non-blocking event-driven concurrency models, with custom message framing and performance monitoring.

Macro Java Compiler & Optimization

- Developed compiler stages spanning lexical analysis, parsing, type checking, IR generation and lowering, register allocation, and MIPS code generation, including stack management and register spilling.
- Implemented compiler analyses and optimization passes including control-flow graph construction, constant propagation, call-graph and class-hierarchy analysis, and function inlining for a Java-like language.

LANGUAGES AND TOOLS

C, C++, Java, Python, Cuda, HDL Verilog, RISC-V Assembly, OCaml, Prolog, MySQL, HTML, CSS, JavaScript

RELEVANT COURSEWORK

Computer Science

Data Structures and Algorithms, Computer Organization and Architecture, Object Oriented Programming, Operating Systems, Paradigms of Programming, Computer Networks, Pattern Recognition and Machine Learning, Advanced Compiler Design, GPU Programming (Ongoing), Foundations of Cryptography (Ongoing)

Mathematics and Economics

Basic Graph Theory, Series and Matrices, Probability & Statistics, Fundamentals of Operations Research, Principles of Economics

EDUCATION

B.Tech., Computer Science

Indian Institute of Technology Madras
2023 – 2027 | CGPA: 9.86

XII – BIEAP

Sri Chaitanya Junior College
2023 | 97.35%

X – AISSE

Sri Chaitanya Techno School
2021 | 95%

ACHIEVEMENTS

Scholastic

- Sri K Krishnamurthi award¹
- JEE Advanced 2023 | AIR 140
- JEE Mains 2023 | AIR 121
- KVPY SA 2022 | AIR 360
- TS EAMCET 2023 | AIR 32
- AP EAPCET 2023 | AIR 117
- NSE 2022 Physics | Top 1% India
- NSE 2022 Chemistry | Top 1% India

Programming

- [LeetCode Knight](#) | 1897
- [Codeforces Expert](#) | 1641
- [CodeChef 3-star](#) | 1624
- Rank 112 | CodeChef Starters 190

POSITIONS OF RESPONSIBILITY

Strategist, Programming Club

CFI, IIT Madras

May 2025 – Apr 2026

- Led and planned events, sessions and contests of the club.
- Contributed to the planning and conduction of sponsored events such as [Bitshift 2026](#) and [Dead-lock 2026](#)
- Conducted software development upskilling activities for Coordinators.

Non-Technical PoRs

- Student rep. CS DCC – A part of the Department Consultative Committee where discussions about departmental welfare take place. | Jun 2024 (Ongoing)
- Class Representative | Aug 2023 – Aug 2024
- Events Coordinator, Computer Science Association | Jan 2025 – May 2025

EXTRA CURRICULAR ACHIEVEMENTS

Trinity College of Music: Level 2 Certificate in Graded Examination in Music Performance, Grade 4 Piano with Merit.

¹ Awarded for securing the 2nd highest CGPA in first year of B.Tech. and Dual Degree.